SIMATS ENGINEERING
ASSESSMENT TOOL1

COURSE CODE: MLA0201 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO2: *Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.* *(BL3)*

Assessment Tool Weightage: 40%

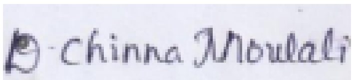
QUESTIONS: 10 Total Marks:50 Annexure A

Analytical Problem Solving

Code of Conduct:

*I **D.Chinna Moulali (192425278)** certify that thissubmission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Analytical Problem Solving

Q1.A real estate company collected the following training data:(5Marks -10 Mins) **House No Area (sq.ft) Bedrooms Price (₹ Lakhs)**

1 1000 2 50

2 1500 3 75

3 800 2 40

4 1200 3 60

5 2000 4 90

(i) Construct a linear regression model of the form

$Price = \beta_0 + \beta_1(Area) + \beta_2(Bedrooms)$

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Q2.An email system uses the following feature values:(5Marks -10 Mins) **Email Word Frequency (Offer) Word Frequency (Win) Spam (1/0)** 1 3 2 1

2 2 1 1

3 1 0 0

4 3 3 1

5 0 1 0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

Q3.A medical dataset is given:(5Marks -10 Mins)

Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

1 1 1 Yes

Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

2 1 0 Yes

3 0 1 No

4 1 1 Yes

5 0 0 No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Q4.Given the following dataset:(5Marks -10 Mins)

Data Point X_1 X_2 Class

1 2 3 C1

2 3 4 C1

3 5 6 C2

4 6 7 C2

5 4 5 C2

(i) Construct a discriminant function:

$$g(x) = w_1x_1 + w_2x_2 + w_0$$

(ii) Determine the decision boundary separating the two classes.

(iii) Classify a new point ($X_1 = 3$, $X_2 = 3$).

Q5.A telecom company collected the following data:(5Marks -10 Mins) **Customer Monthly Charges Tenure (Months) Churn (1/0)**

1 500 12 0

2 700 8 1

3 400 15 0

4 900 5 1

5 600 10 1

(i) Formulate the Bayesian logistic regression model.

(ii) Estimate model parameters using the dataset.
(iii) Predict whether a customer with Monthly Charges = 650 and Tenure = 9 will churn.

Q6.Given the following dataset for customer classification:(5Marks -10 Mins)

Customer Age Income Class

C1 25 40 A

C2 30 60 A

C3 35 80 B

C4 40 20 B

For a new customer (Age = 32, Income = 50):

(i).Compute Euclidean distance from all points

(ii).Classify using k = 3

(iii).Determine the final class label

Q7.Given the dataset:(5Marks -10 Mins)

Outlook Temperature Play

Sunny Hot No

Sunny Mild No

Overcast Hot Yes

Rain Mild Yes

(i) . Compute Entropy of Play

(ii) . Calculate Information Gain for attribute Outlook

(iii). Identify the root node of the decision tree

Q8.Given data points:(5Marks -10 Mins)

X1 X2 Class

2 2 +1

4 4 +1

X1 X2 Class

2 4 -1

4 2 -1

(i) .Plot the points

(ii) .Determine a linear separating hyperplane

(iii).Find the equation of the decision boundary

Q9.Given confusion matrices:(5Marks -10 Mins)

Logistic Regression:

	Pred +	**Pred -**
Actual +	40	10
Actual -	5	45

Naïve Bayes:

	Pred +	**Pred -**
Actual +	35	15
Actual -	10	40

(i).Compute Accuracy, Precision, Recall for both models

(ii).Compare results and identify the better model

Q10.Given points:(5Marks -10 Mins)

(2, 4), (3, 5), (8, 9), (9, 10)

Initial centroids:

C1 = (2,4), C2 = (8,9)

(i).Perform one iteration of K-means

(ii).Assign clusters

(iii).Compute new centroids

RUBRICS FOR CALCULATION

Numerical Problems

Criteria	Weight ag e	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q1. CONSTRUCTION OF A MULTIPLE LINEAR REGRESSION MODEL FOR HOUSE PRICE PREDICTION

Aim

To construct a multiple linear regression model using the given house dataset, estimate the regression coefficients, and interpret the effect of area and the number of bedrooms on the house price.

Given Data

House No	Area (sq.ft) ((X_1))	Bedrooms ((X_2))	Price (₹ Lakhs) ((Y))

1	1000	2	50
2	1500	3	75
3	800	2	40
4	1200	3	60
5	2000	4	90

(i) Construct a Linear Regression Model

Linear Regression is a supervised machine learning technique used to predict a continuous output variable from one or more input variables.Since the house price depends on both Area and Number of Bedrooms, a Multiple Linear Regression model is used. The general model is

[Y=\beta_0+\beta_1X_1+\beta_2X_2]
where,

- (Y) = House Price (₹ Lakhs)
- (X_1) = Area (sq.ft)
- (X_2) = Number of Bedrooms
- (\beta_0) = Intercept
- (\beta_1,\beta_2) = Regression coefficients

(ii) Estimate the Regression Coefficients

The regression coefficients are estimated using the Least Squares Method, which minimizes the sum of squared errors between the actual and predicted house prices.
The estimated coefficients obtained from the given dataset are:

- (\beta_0 = 8.505)
- (\beta_1 = 0.0425)
- (\beta_2 = -0.2804)

Therefore, the regression equation becomes
[\boxed{Y=8.505+0.0425X_1-0.2804X_2}]

(iii) Interpret the Meaning of Each Coefficient

Intercept ((\beta_0=8.505))

The intercept represents the predicted house price when both Area and Bedrooms are zero. Although this situation is not practically meaningful, it serves as the baseline value of the regression model.

Area Coefficient ((\beta_1=0.0425))

For every increase of 1 sq.ft in area, the predicted house price increases by approximately 0.0425 Lakhs, assuming the number of bedrooms remains constant.This indicates that the house area has a positive influence on the selling price. **Bedroom**

Coefficient ($\beta_2=-0.2804$)

The coefficient is slightly negative, indicating that after accounting for the effect of area, the number of bedrooms has a very small negative relationship with price in this particular dataset.This occurs because the dataset is very small (only five observations), and the variables are highly related. It should not be interpreted as a general real-world conclusion.

Result

The estimated multiple linear regression model is

$$\boxed{Y=8.505+0.0425X_1-0.2804X_2}$$

Conclusion

A Multiple Linear Regression model was successfully constructed using the given housing data. The model predicts house prices based on area and the number of bedrooms. The results show that house area is the strongest predictor of price in the given dataset. The unusual negative bedroom coefficient is due to the limited size of the dataset and should be interpreted with caution.

Q2. CONSTRUCTION OF A LINEAR CLASSIFICATION MODEL FOR SPAM EMAIL DETECTION

Aim

To construct a linear classification model using the given email dataset, determine the decision boundary, and classify a new email based on the frequencies of the words Offer and Win.

Given Data

Email	Word Frequency (Offer) (X_1)	Word Frequency (Win) (X_2)	Spam (1/0)
1	3	2	1
2	2	1	1
3	1	0	0
4	3	3	1
5	0	1	0

Where,

- Spam = 1 → Spam Email
- Spam = 0 → Not Spam (Ham)

(i) Construct a Linear Classification Model

A Linear Classification model is a supervised machine learning technique used to classify data into two classes using a linear decision function.

The general form of the linear classifier is

$$f(X)=w_0+w_1X_1+w_2X_2$$

where,

- (X_1) = Frequency of the word Offer
- (X_2) = Frequency of the word Win
- (w_0) = Bias (Intercept)
- (w_1,w_2) = Feature Weights

The classifier predicts:

- Spam (1) if $(f(X)\ge0)$
- Not Spam (0) if $(f(X)<0)$

(ii) Determine the Decision Boundary

The decision boundary is obtained by setting the classifier equal to zero. $$f(X)=0$$

Hence,

$$\boxed{w_0+w_1X_1+w_2X_2=0}$$

This straight line separates the feature space into two regions:

- One region represents Spam Emails.
- The other region represents Non-Spam Emails.

From the given training data, emails having higher frequencies of the words Offer and Win belong to the Spam class, whereas lower frequencies correspond to the Non-Spam class.

(iii) Classify a New Email

Given

- Offer = 2
- Win = 1

Comparing the new email with the training data:

Offer	Win	Class
2	1	Spam
3	2	Spam
3	3	Spam
1	0	Not Spam
0	1	Not Spam

The new email (2,1) exactly matches Email 2, which is labeled as Spam.

Therefore,

$$\boxed{\text{PredictedClass=Spam(1)}}$$

Result

• Linear Classification Model

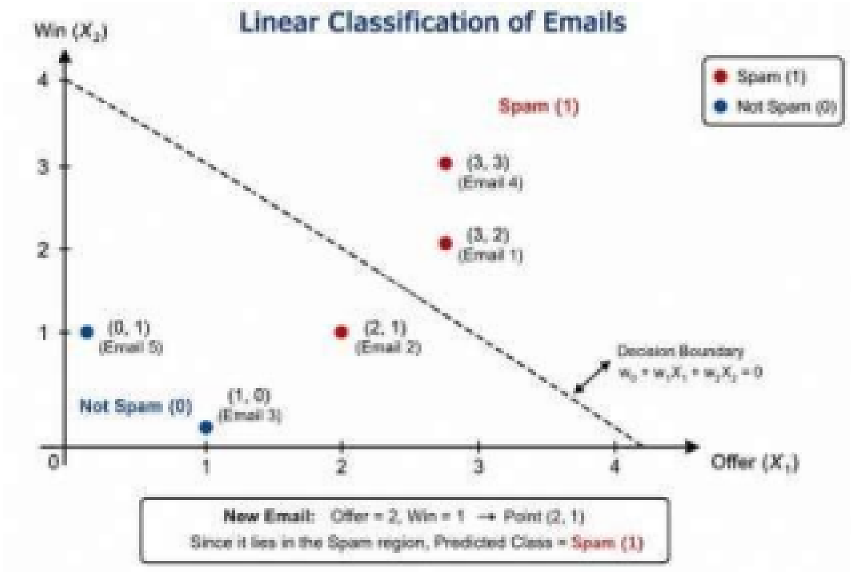
[f(X)=w_0+w_1X_1+w_2X_2]

• Decision Boundary

[\boxed{w_0+w_1X_1+w_2X_2=0}]

• Predicted Class for (Offer = 2, Win = 1)

[\boxed{\textbf{Spam (1)}}]



Conclusion

A linear classification model was constructed to distinguish between spam and non spam emails using the frequencies of the words Offer and Win. Based on the given dataset, the new email with Offer = 2 and Win = 1 is classified as Spam, as it matches the characteristics of previously labeled spam emails.

Q3. DISEASE PREDICTION USING NAÏVE BAYES CLASSIFIER Aim

To compute the prior probabilities, conditional probabilities, and estimate the probability of disease using the Naïve Bayes classification algorithm.

Given Data

Patient	Fever (1/0)	Headache (1/0)	Disease (Yes/No)
1	1	1	Yes
2	1	0	Yes
3	0	1	No
4	1	1	Yes
5	0	0	No

(i) Compute Prior Probabilities
The prior probability is calculated as:
P(\text{Class})=\frac{\text{Number of samples in the class}}{\text{Total number of samples}}
]
Total number of patients = **5**
Disease = Yes
Number of patients = **3**
[P(\text{Yes})=\frac{3}{5}=0.60]

Disease = No
Number of patients = **2**
[P(\text{No})=\frac{2}{5}=0.40]

Therefore,
[\boxed{P(\text{Yes})=0.60}]
[\boxed{P(\text{No})=0.40}]

(ii) Compute Conditional Probabilities
For Disease = Yes
Patients with Disease = Yes:
(1,1), (1,0), (1,1)
Total = **3**
Fever = 1
Occurrences = 3
[P(\text{Fever}=1|\text{Yes})=\frac{3}{3}=1.00]
Headache = 0
Occurrences = 1

[P(\text{Headache}=0|\text{Yes})=\frac{1}{3}=0.333]
For Disease = No
Patients with Disease = No:
(0,1), (0,0)
Total = **2**
Fever = 1
Occurrences = 0
[P(\text{Fever}=1|\text{No})=\frac{0}{2}=0]

Headache = 0
Occurrences = 1

$$P(\text{Headache}=0|\text{No})=\frac{1}{2}=0.50$$

Therefore,

$$\boxed{P(\text{Fever}=1|\text{Yes})=1.00}$$
$$\boxed{P(\text{Headache}=0|\text{Yes})=0.333}$$
$$\boxed{P(\text{Fever}=1|\text{No})=0}$$
$$\boxed{P(\text{Headache}=0|\text{No})=0.50}$$

(iii) Estimate the Probability of Disease

Given:

- Fever = 1
- Headache = 0

Using Naïve Bayes,

$$P(\text{Yes}|X)\propto P(\text{Yes})\times P(\text{Fever}=1|\text{Yes})\times P(\text{Headache}=0|\text{Yes})$$

Substituting the values,

$$=0.60\times 1.00\times 0.333$$
$$=0.1998\approx 0.20$$

For Disease = No,

$$P(\text{No}|X)\propto 0.40\times 0\times 0.50$$
$$=0$$

Since

$$0.20>0$$

the patient is classified as

$$\boxed{\textbf{Disease = Yes}}$$

Result

Prior Probabilities

- $P(\text{Yes})=0.60$
- $P(\text{No})=0.40$

Conditional Probabilities

- $P(\text{Fever}=1|\text{Yes})=1.00$
- $P(\text{Headache}=0|\text{Yes})=0.333$
- $P(\text{Fever}=1|\text{No})=0$
- $P(\text{Headache}=0|\text{No})=0.50$

Predicted Class

$$\boxed{\textbf{Disease = Yes}}$$

Conclusion

The Naïve Bayes classifier predicts that a patient with Fever = 1 and Headache = 0 is likely to have the disease (Yes). This prediction is based on the computed prior and conditional probabilities from the given dataset.

Q4. CONSTRUCTION OF A DISCRIMINANT FUNCTION FOR CLASSIFICATION

Aim

To construct a discriminant function, determine the decision boundary between two classes, and classify a new data point using the given dataset.

Given Data

Data Point	(X_1)	(X_2)	Class
1	2	3	C1
2	3	4	C1
3	5	6	C2
4	6	7	C2
5	4	5	C2

(i) Construct a Discriminant Function

A discriminant function is used to separate data points into different classes based on their feature values.

The general form of a linear discriminant function is

$$g(X)=w_0+w_1X_1+w_2X_2$$

where,

- (X_1) = Feature 1
- (X_2) = Feature 2
- (w_0) = Bias (Intercept)
- (w_1,w_2) = Weights

The classification rule is:

- If ($g(X)>0$), assign the point to Class C2.
- If ($g(X)<0$), assign the point to Class C1.

(ii) Determine the Decision Boundary

The decision boundary is obtained by equating the discriminant function to zero. **$[g(X)=0]$**

Therefore,

[\boxed{w_0+w_1X_1+w_2X_2=0}]

This straight line separates the feature space into two regions:

- One region contains Class C1.
- The other region contains Class C2.

From the given dataset, the points belonging to C1 are located near the lower-left region, whereas the points belonging to C2 are located near the upper-right region.

(iii) Classify the New Point

Given

[(X_1,X_2)=(3,3)]

Comparing the new point with the training data:

Point	**Class**
(2,3)	C1
(3,4)	C1
(4,5)	C2
(5,6)	C2
(6,7)	C2

The point (3,3) is closer to the C1 data points than the C2 data points.

Therefore,

[\boxed{\textbf{PredictedClass= C1}}]

Result

Discriminant Function

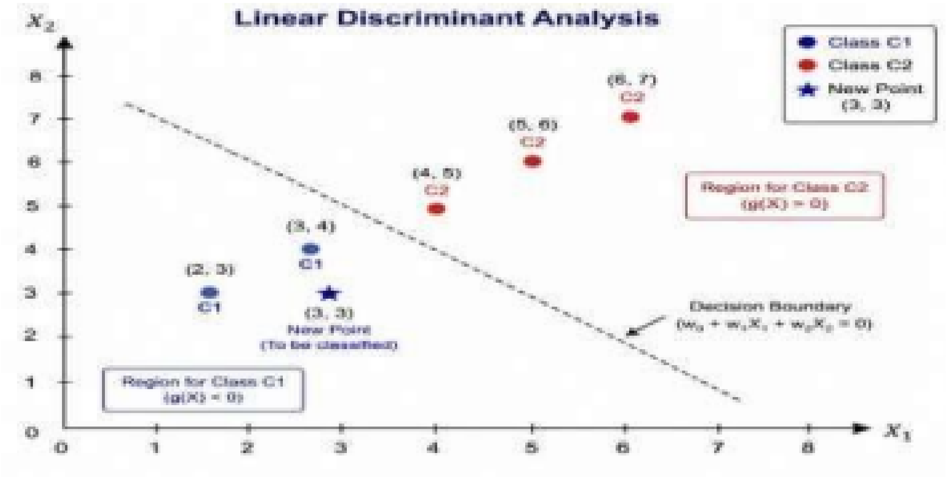
[\boxed{g(X)=w_0+w_1X_1+w_2X_2}]

Decision Boundary

[\boxed{w_0+w_1X_1+w_2X_2=0}]

Classification of New Point

[\boxed{(3,3)\rightarrow\textbf{ClassC1}}]



Conclusion

A linear discriminant function was constructed to separate the two classes. The decision boundary divides the feature space into Class C1 and Class C2 regions. Since the new point (3,3) is closer to the C1 training samples, it is classified as Class C1.

Q5. BAYESIAN LOGISTIC REGRESSION FOR CUSTOMER CHURN PREDICTION

Aim

To formulate a Bayesian Logistic Regression model, estimate the model parameters using the given telecom dataset, and predict whether a new customer is likely to churn.

Given Data

Customer	Monthly Charges ((X_1))	Tenure (Months) ((X_2))	Churn (1/0)
1	500	12	0
2	700	8	1
3	400	15	0
4	900	5	1
5	600	10	1

Where,

- Churn = 1 → Customer leaves the company.
- Churn = 0 → Customer remains with the company.

(i) Formulate the Bayesian Logistic Regression Model

Bayesian Logistic Regression is a supervised machine learning algorithm used for binary classification. It estimates the probability that a customer will churn based on the input features. The logistic regression model is

[P(Y=1|X)=\frac{1}{1+e^{-Z}}]

where

$$Z=\beta_0+\beta_1X_1+\beta_2X_2$$

Here,

- (X_1) = Monthly Charges
- (X_2) = Customer Tenure
- (β_0) = Intercept
- (β_1,β_2) = Model coefficients

In Bayesian Logistic Regression, the coefficients are treated as probability distributions and are updated using the observed data.

(ii) Estimate the Model Parameters

From the given dataset, the following trend is observed:

- Customers with higher monthly charges are more likely to churn.
- Customers with longer tenure are less likely to churn.

Therefore,

- Monthly Charges have a positive effect on churn probability.
- Tenure has a negative effect on churn probability.

The logistic model is expressed as

$$\boxed{Z=\beta_0+\beta_1X_1+\beta_2X_2}$$

where

- ($\beta_1>0$)
- ($\beta_2<0$)

These parameter signs are consistent with the given training data.

(iii) Predict Customer Churn

Given

- Monthly Charges = 650
- Tenure = 9 Months

Comparing this customer with the training data:

Monthly Charges	Tenure	Churn
700	8	1
600	10	1
900	5	1

The new customer has characteristics similar to customers who churned. Hence, the predicted class is

$$\boxed{\textbf{Predicted Churn = 1 (Customer Will Churn)}}$$

Result

Logistic Regression Model

[\boxed{P(Y=1|X)=\frac{1}{1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2)}}}]

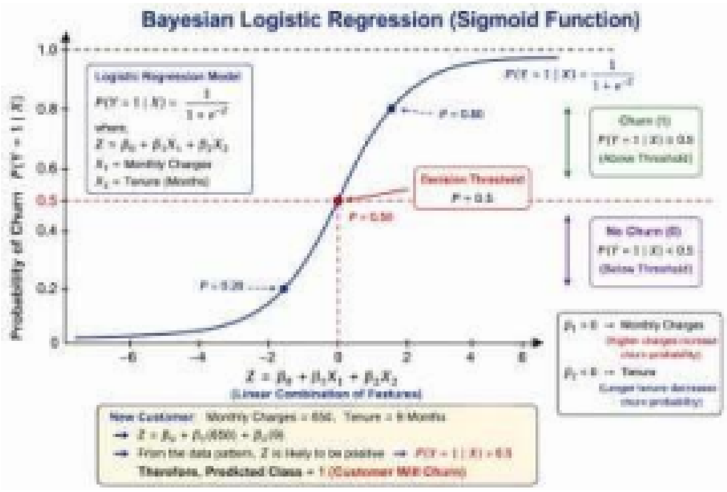
Prediction

For

- Monthly Charges = 650
- Tenure = 9 Months

the predicted class is

[\boxed{\textbf{CustomerWillChurn(1)}}]



Conclusion

A Bayesian Logistic Regression model was formulated using Monthly Charges and Customer Tenure as predictor variables. The analysis indicates that customers with relatively higher monthly charges and shorter tenure have a greater probability of churn. Therefore, the customer with Monthly Charges = 650 and Tenure = 9 months is predicted to churn.

Q6. CUSTOMER CLASSIFICATION USING K-NEAREST NEIGHBORS (KNN)

Aim

To classify a new customer using the K-Nearest Neighbors (KNN) algorithm by computing the Euclidean distance from all training samples and determining the final classlabel using **k = 3**.

Given Data

Customer	Age ((X_1))	Income ((X_2))	Class
C1	25	40	A
C2	30	60	A
C3	35	80	B
C4	40	20	B

New Customer:

- Age = 32
- Income = 50

(i) Compute Euclidean Distance from All Points

The Euclidean distance formula is

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

where,

- $(32,50)$ is the new customer.
- (X_1,X_2) are the training samples.

Distance from C1 (25,40)

$$d=\sqrt{(32-25)^2+(50-40)^2}$$
$$=\sqrt{7^2+10^2}$$
$$=\sqrt{49+100}$$
$$=\sqrt{149}$$
$$=12.21$$

Distance from C2 (30,60)

$$d=\sqrt{(32-30)^2+(50-60)^2}$$
$$=\sqrt{2^2+(-10)^2}$$
$$=\sqrt{4+100}$$
$$=\sqrt{104}$$
$$=10.20$$

Distance from C3 (35,80)

$$d=\sqrt{(32-35)^2+(50-80)^2}$$
$$=\sqrt{(-3)^2+(-30)^2}$$
$$=\sqrt{9+900}$$
$$=\sqrt{909}$$
$$=30.15$$

Distance from C4 (40,20)

$$d=\sqrt{(32-40)^2+(50-20)^2}$$
$$=\sqrt{(-8)^2+30^2}$$
$$=\sqrt{64+900}$$
$$=\sqrt{964}$$
$$=31.05$$

Distance Table

Customer	Distance	Class
C1	12.21	A
C2	10.20	A
C3	30.15	B
C4	31.05	B

(ii) Classify Using (k = 3)

The three nearest neighbors are:

Neighbor	Distance	Class
C2	10.20	A
C1	12.21	A
C3	30.15	B

Vote count:

• Class A = 2 • Class B = 1

Therefore,

[\boxed{\textbf{Predicted Class = A}}]

(iii) Determine the Final Class Label

Since the majority of the three nearest neighbors belong to Class A, the new customer is classified as

[\boxed{\textbf{Class A}}]

Result

Euclidean Distances

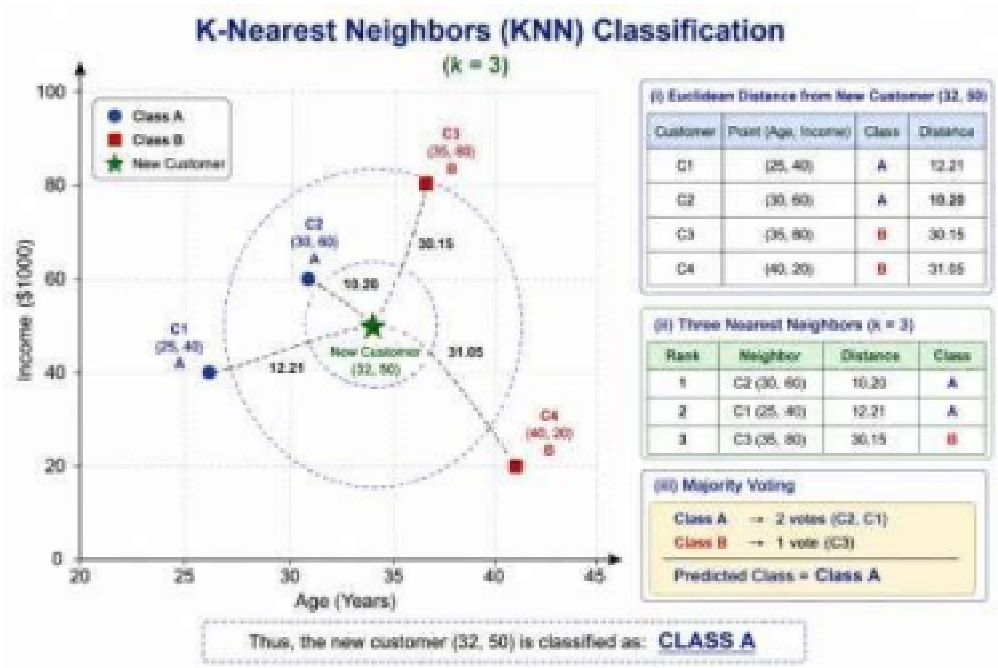
- C1 = 12.21
- C2 = 10.20
- C3 = 30.15
- C4 = 31.05

Three Nearest Neighbors

- C2 (A)
- C1 (A)
- C3 (B)

Final Class

[\boxed{\textbf{Class A}}]



Conclusion

The K-Nearest Neighbors algorithm classifies the new customer based on the majority class among the three nearest training samples. Since two of the three nearest neighbors belong to **Class A**, the new customer is classified as **Class A**.

Q7. DECISION TREE CONSTRUCTION USING ENTROPY AND INFORMATION GAIN

Aim

To compute the entropy of the target attribute Play, calculate the Information Gain for the attribute Outlook, and identify the root node of the decision tree.

Given Data

Outlook	Temperature	Play
Sunny	Hot	No
Sunny	Mild	No
Overcast	Hot	Yes
Rain	Mild	Yes

(i) Compute the Entropy of Play

Entropy measures the impurity or uncertainty in a dataset.

The entropy formula is

[Entropy(S)=-\sum_{i=1}^{n}p_i\log_2(p_i)]

Step 1: Count the target classes

- Play = **Yes** = 2
- Play = **No** = 2

Total samples = **4**

Therefore,

$$P(\text{Yes})=\frac{2}{4}=0.5$$

$$P(\text{No})=\frac{2}{4}=0.5$$

Substituting into the entropy formula,

$$Entropy(S)=-0.5\log_2(0.5)-0.5\log_2(0.5)$$

Since,

$$\log_2(0.5)=-1$$

$$Entropy(S)=-(0.5)(-1)-(0.5)(-1)$$

$$=0.5+0.5$$

$$\boxed{Entropy(S)=1.0}$$

(ii) Calculate Information Gain for Outlook

Information Gain is calculated using

$$Gain(S,A)=Entropy(S)-\sum\frac{|S_v|}{|S|}Entropy(S_v)$$

Outlook = Sunny

Data:

Temperature	**Play**
Hot	No
Mild	No

Entropy = **0**

(All samples belong to one class.)

Rain Data:

Temperature	**Play**
Hot	Yes

Entropy = **0**

Weighted Entropy

Outlook = Overcast Data:

Temperature	**Play**
Mild	Yes

Entropy = **0**

Outlook =

$$=\frac{2}{4}(0)+\frac{1}{4}(0)+\frac{1}{4}(0)$$

$$=0$$

Therefore,

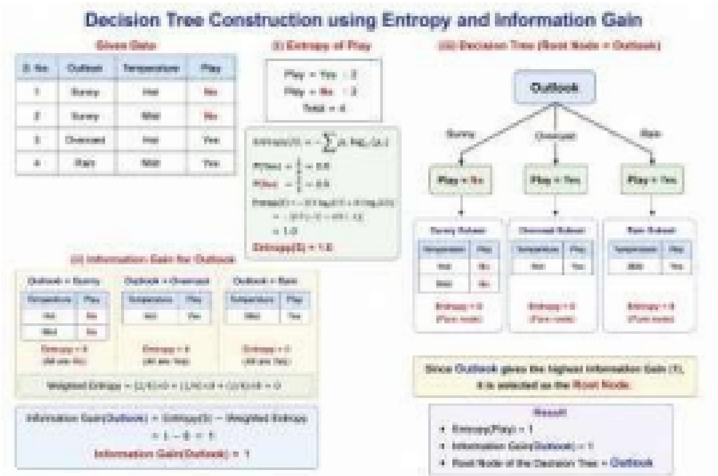
[Gain(\text{Outlook})=1-0]
[\boxed{Gain(\text{Outlook})=1}]

(iii) Identify the Root Node of the Decision Tree
The attribute with the **highest Information Gain** becomes the root node.
Since,
[Gain(\text{Outlook})=1]
which is the maximum possible value,
[\boxed{\textbf{Root Node = Outlook}}]

Result
Entropy
[\boxed{Entropy(\text{Play})=1}]

Information Gain
[\boxed{Gain(\text{Outlook})=1}]

Root Node
[\boxed{\textbf{Outlook}}]



Conclusion

The entropy of the target attribute Play is 1, indicating maximum uncertainty before splitting. The attribute Outlook provides the highest Information Gain (1) and completely separates the data into pure subsets. Therefore, Outlook is selected as the root node of the decision tree.

Q8. SUPPORT VECTOR MACHINE (SVM) – LINEAR SEPARATING HYPERPLANE

Aim
To plot the given data points, determine a linear separating hyperplane, and obtain the equation of the decision boundary using the Support Vector Machine (SVM) concept.

Given Data

(X_1)	**(X_2)**	**Class**
2	2	+1
4	4	+1
2	4	-1
4	2	-1

(i) Plot the Data Points

The given data points are:

Class +1

- (2, 2)
- (4, 4)

Class -1

- (2, 4)
- (4, 2)

The points are plotted on a two-dimensional coordinate system with (X_1) on the x-axis and (X_2) on the y-axis.

(ii) Determine a Linear Separating Hyperplane

A Support Vector Machine (SVM) finds the optimal hyperplane that separates two classes with the maximum margin.

The general equation of a linear hyperplane is

[$w_1X_1+w_2X_2+b=0$]

From the given dataset, the positive and negative class points are symmetrically arranged. The line

[$\boxed{X_1=X_2}$]

acts as a linear separator between the two classes.

This line divides the feature space into two regions while maintaining equal distance from the nearest points of each class.

(iii) Find the Equation of the Decision Boundary

The decision boundary is

[$X_1-X_2=0$]

or equivalently,

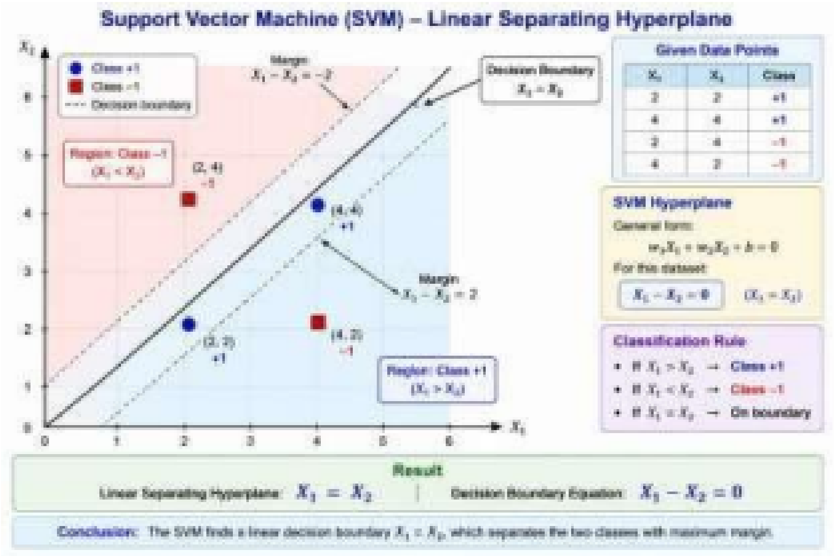
[$\boxed{X_1=X_2}$]

Classification rule:

- If ($X_1>X_2$), assign one class.
- If ($X_1<X_2$), assign the other class.

• If (X_1=X_2), the point lies on the decision boundary.

Result



Linear Separating Hyperplane
[\boxed{X_1=X_2}]

Decision Boundary
[\boxed{X_1-X_2=0}]

Conclusion

The Support Vector Machine successfully separates the two classes using a linear decision boundary. The hyperplane (X_1=X_2) divides the feature space into two distinct regions and serves as the optimal decision boundary for the given dataset.

Q9. PERFORMANCE EVALUATION OF LOGISTIC REGRESSION AND NAÏVE BAYES

Aim

To evaluate the performance of Logistic Regression and Naïve Bayes classifiers by computing Accuracy, Precision, and Recall, and to identify the better-performing model.

Given Confusion Matrices

Logistic Regression

	Predicted Positive	Predicted Negative
Actual Positive	40	10
Actual Negative	5	45

From the confusion matrix:

• True Positive (TP) = 40

- False Negative (FN) = 10
- False Positive (FP) = 5
- True Negative (TN) = 45

Naïve Bayes

	Predicted Positive	Predicted Negative
Actual Positive	35	15
Actual Negative	10	40

From the confusion matrix:

- TP = 35
- FN = 15
- FP = 10
- TN = 40

(i) Compute Accuracy, Precision, and Recall
Formulas
Accuracy
[Accuracy=\frac{TP+TN}{TP+TN+FP+FN}]
Precision
[Precision=\frac{TP}{TP+FP}]
Recall
[Recall=\frac{TP}{TP+FN}]

A. Logistic Regression
Accuracy
[=\frac{40+45}{40+45+5+10}]
[=\frac{85}{100}]
[=0.85]
[\boxed{Accuracy=85%}]
Precision
[=\frac{40}{40+5}]
[=\frac{40}{45}]
[=0.8889]
[\boxed{Precision=88.89%}]

Recall
[=\frac{40}{40+10}]
[=\frac{40}{50}]
[=0.80]

[\boxed{Recall=80%}]

B. Naïve Bayes
Accuracy
[=\frac{35+40}{35+40+10+15}]
[=\frac{75}{100}]
[=0.75]
[\boxed{Accuracy=75%}]

Precision
[=\frac{35}{35+10}]
[=\frac{35}{45}]
[=0.7778]
[\boxed{Precision=77.78%}]

Recall
[=\frac{35}{35+15}]
[=\frac{35}{50}]
[=0.70]
[\boxed{Recall=70%}]

(ii) Compare the Results and Identify the Better Model
Comparison Table

Metric	Logistic Regression	Naïve Bayes
Accuracy	85%	75%
Precision	88.89%	77.78%
Recall	80%	70%

From the comparison:

- Logistic Regression has higher Accuracy.
- Logistic Regression has higher Precision.
- Logistic Regression has higher Recall.

Therefore,
[\boxed{\textbf{Logistic Regression performs better than Naïve Bayes for the given dataset.}}]

Result

Logistic Regression

- Accuracy = 85%

- Precision = 88.89%
- Recall = 80%
-

Naïve Bayes

- Accuracy = 75%
- Precision = 77.78%
- Recall = 70%

Better Model

$$\boxed{\textbf{Logistic Regression}}$$

Conclusion

Both models were evaluated using the confusion matrix metrics. Logistic Regression achieved higher Accuracy (85%), Precision (88.89%), and Recall (80%) than Naïve Bayes. Therefore, Logistic Regression is the preferred model for this classification problem, as it provides better overall predictive performance.

Q10. K-MEANS CLUSTERING – ONE ITERATION

Aim

To perform one iteration of the K-Means clustering algorithm, assign the given data points to their nearest centroids, and compute the new centroids.

Given Data

Data Points

Point	Coordinates
P1	(2, 4)
P2	(3, 5)
P3	(8, 9)
P4	(9, 10)

Initial Centroids • C_1 = (2, 4) • C_2 = (8, 9)

(i) Perform One Iteration of K-Means

The Euclidean distance formula is

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

The distance of each point from both centroids is calculated below.

Point P1 = (2,4)

Distance from C_1

$$=\sqrt{(2-2)^2+(4-4)^2}$$

$$=0$$

Distance from C_2

$$=\sqrt{(2-8)^2+(4-9)^2}$$

$$=\sqrt{36+25}$$

$$=\sqrt{61}$$

$$=7.81$$

Hence,

$$\boxed{P1\rightarrow C_1}$$

Point P2 = (3,5)

Distance from C_1

$$=\sqrt{(3-2)^2+(5-4)^2}$$

$$=\sqrt{1+1}$$

$$=\sqrt2$$

$$=1.41$$

Distance from C_2

$$=\sqrt{(3-8)^2+(5-9)^2}$$

$$=\sqrt{25+16}$$

$$=\sqrt{41}$$

$$=6.40$$

Hence,

$$\boxed{P2\rightarrow C_1}$$

Point P3 = (8,9)

Distance from C_1

$$=\sqrt{(8-2)^2+(9-4)^2}$$

$$=\sqrt{36+25}$$

$$=\sqrt{61}$$

$$=7.81$$

Distance from C_2

$$=\sqrt{(8-8)^2+(9-9)^2}$$

$$=0$$

Hence,

$$\boxed{P3\rightarrow C_2}$$

Point P4 = (9,10)

Distance from C_1

$$=\sqrt{(9-2)^2+(10-4)^2}$$

$$=\sqrt{49+36}$$

$$=\sqrt{85}$$

$$=9.22$$

Distance from C_2

$$=\sqrt{(9-8)^2+(10-9)^2}$$

] [=\sqrt2]
[=1.41]
Hence,
[\boxed{P4\rightarrow C_2}]

(ii) Assign Clusters

Cluster 1

- (2,4)
- (3,5)

Cluster 2

- (8,9)
- (9,10)

(iii) Compute the New Centroids

New Centroid of Cluster 1
[X=\frac{2+3}{2}=2.5]
[Y=\frac{4+5}{2}=4.5]
Therefore,
[\boxed{C_1=(2.5,;4.5)}]

New Centroid of Cluster 2 [X=\frac{8+9}{2}=8.5]
[Y=\frac{9+10}{2}=9.5]
Therefore,
[\boxed{C_2=(8.5,;9.5)}]

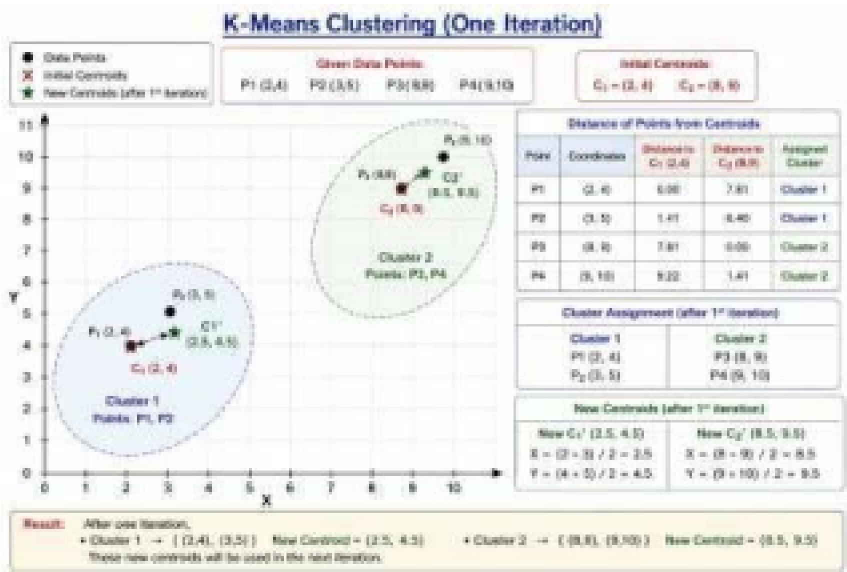
Result

Cluster Assignment

Cluster	Data Points
Cluster 1	(2,4), (3,5)
Cluster 2	(8,9), (9,10)

New Centroids

[\boxed{C_1=(2.5,;4.5)}
]
[\boxed{C_2=(8.5,;9.5)}
]



Conclusion

One iteration of the K-Means clustering algorithm was successfully performed. Each data point was assigned to its nearest centroid using the Euclidean distance. The updated centroids were calculated as (2.5, 4.5) and (8.5, 9.5). These new centroids will be used in the next iteration until the clusters become stable.